

Balancing an Inverted Pendulum with a Controlled Reaction Wheel

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INTRODUCTION

System Overview

- The system is comprised of an arm to be held in its upright, vertical position
- Reaction wheel attached to arm is spun by motor
- Balanced by the wheel's angular momentum

Applications

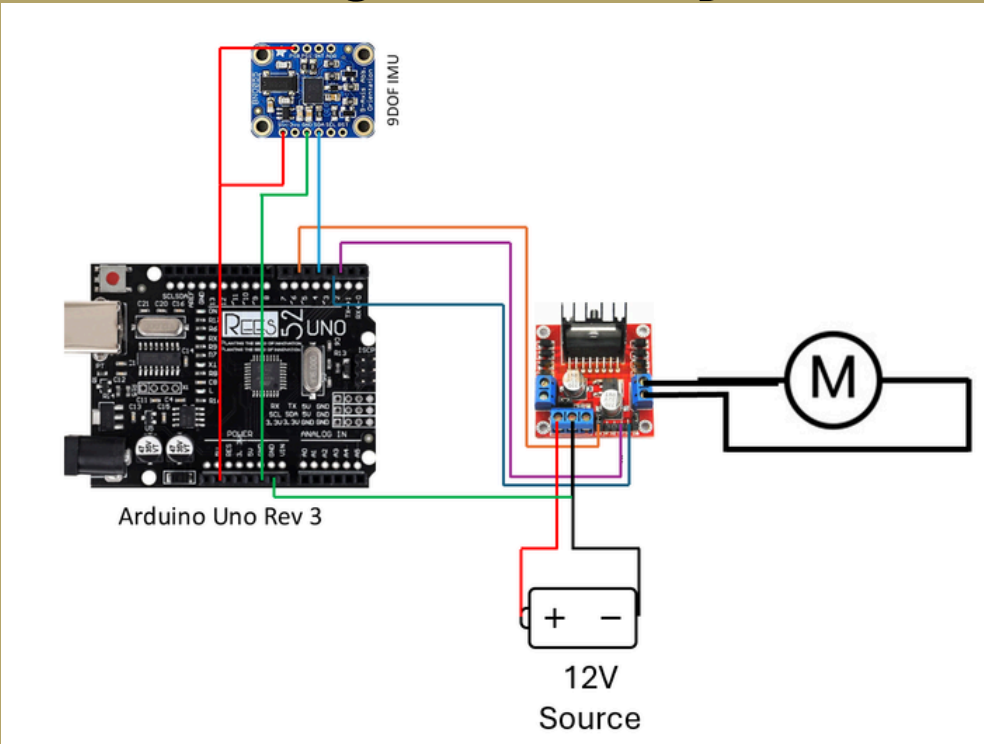
- Satellites use wheels to control attitude
- Reaction wheels could balance a robot limb
- Reaction wheel could stabilize camera view

HARDWARE SETUP

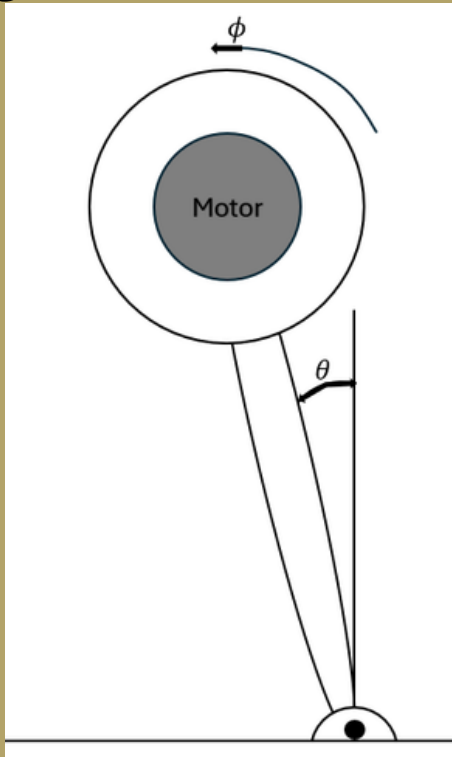
Instrumentation, Sensors and Inputs

- Arduino Uno Rev 3 SMD
 - Inputs: Gains and upright commands
- Adafruit 9-DOF Orientation IMU
 - Measures acceleration in all 3 directions
- 200 RPM Small 3V DC motor and H-Bridge
 - Measures wheel position & speed
- 3D Printed Stand, Wheel, and Arm

Circuit Diagram and System Diagram

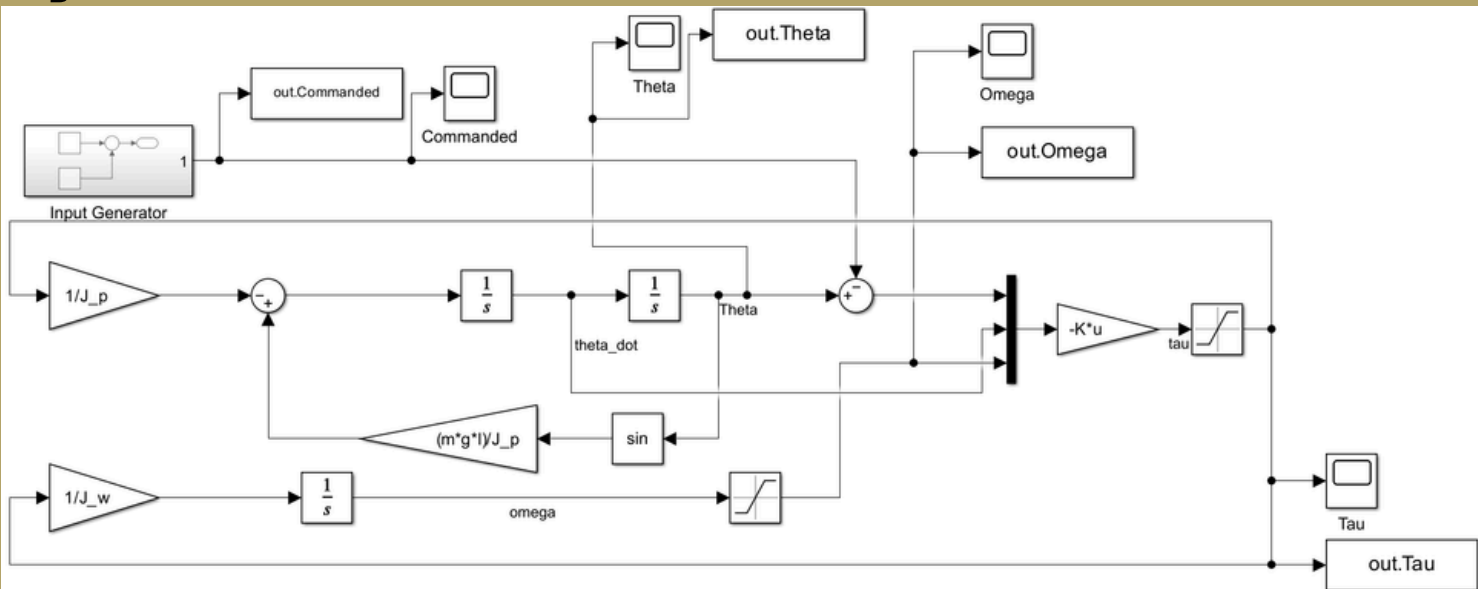


Circuit Diagram for the controller and sensor system



System Diagram

System Model and State Matrix



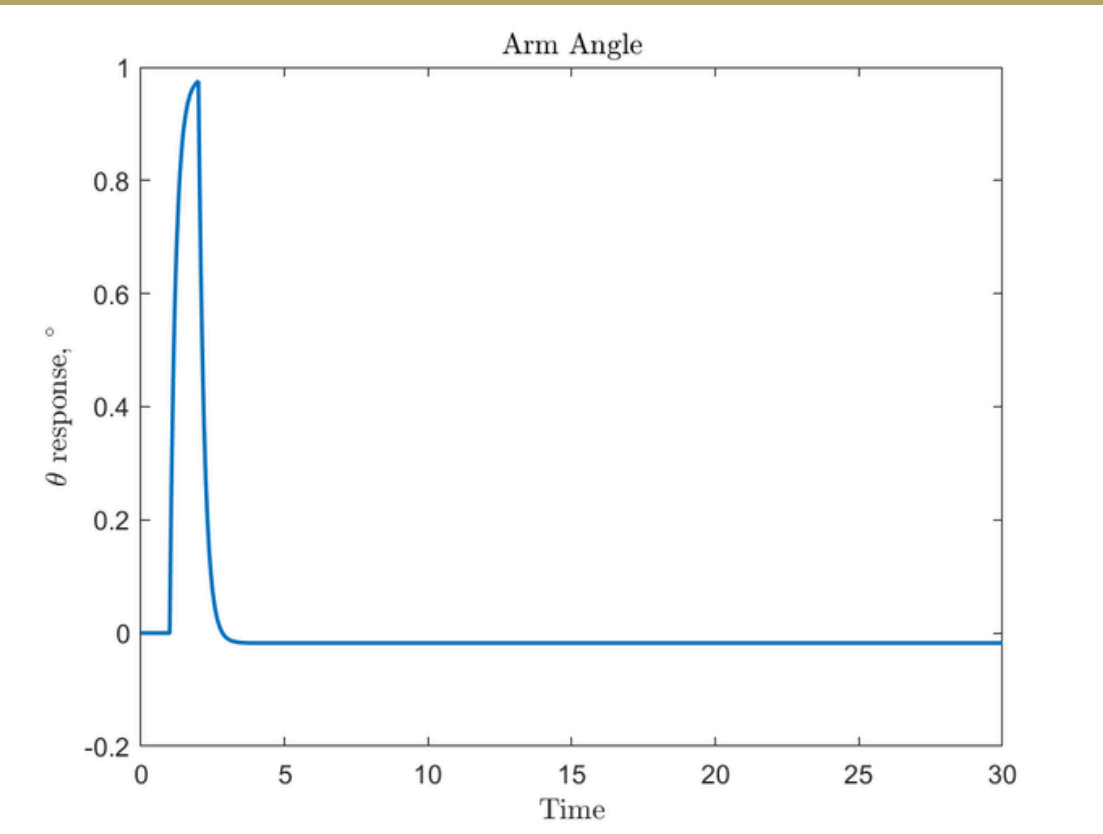
Simulink Model for Reaction Flywheel System

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ \frac{mgl}{J_p} & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{1}{J_p} \\ \frac{1}{J_w} \end{bmatrix} u$$

State space matrix representation of system.

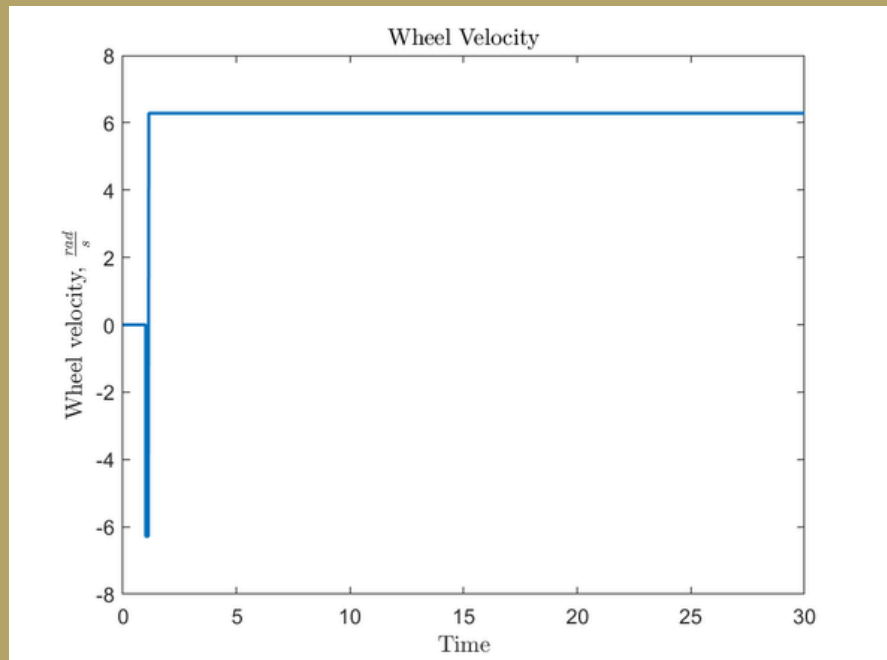
EVALUATION AND RESULTS

SIMULINK Model Results:

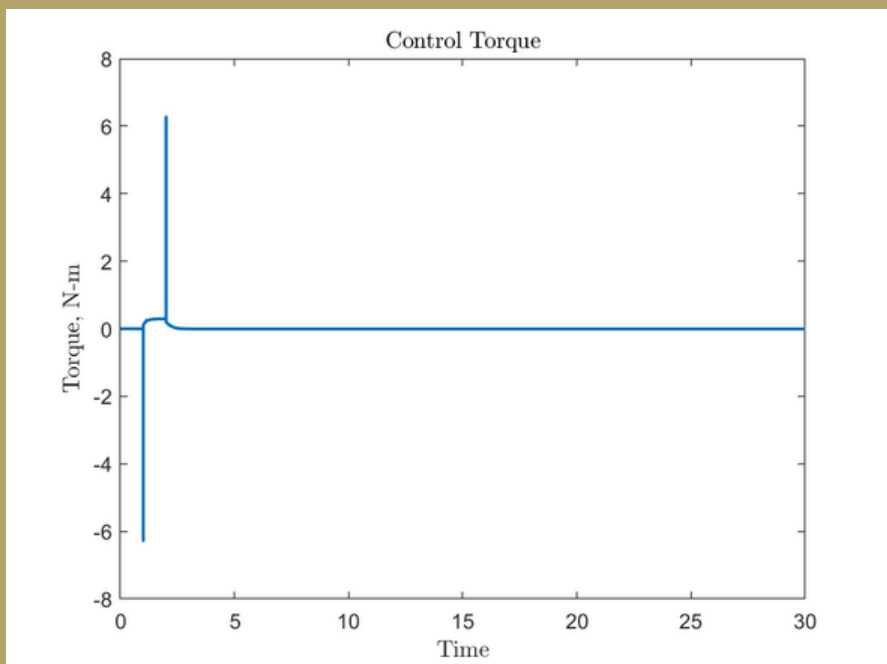


Arm Angle following a 1 degree disturbance

The simulink model was commanded to 0 degrees, and an impulse of 1 degree was applied at 1 second. The response is detailed in these graphs, the system was able to recover.

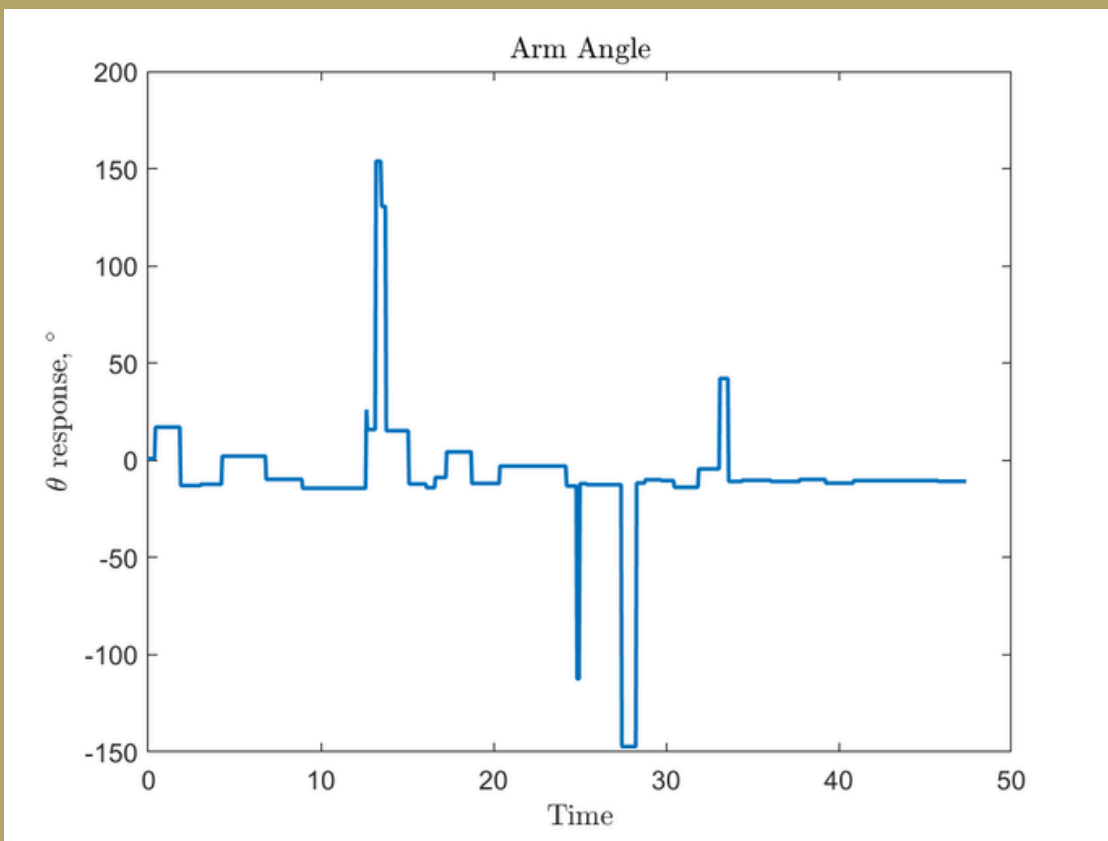


Response of wheel velocity when given 1 deg impulse.



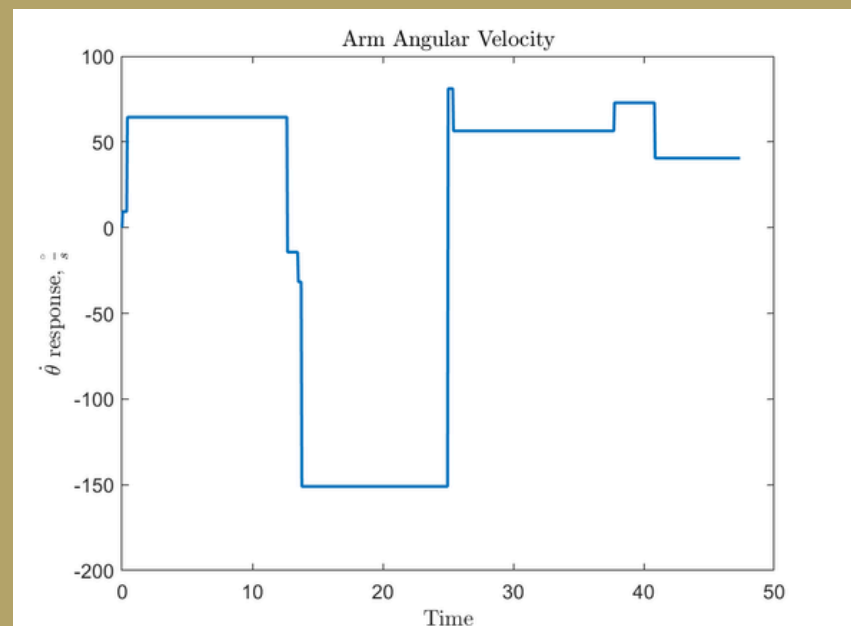
Response of torque when given 1 deg impulse.

Experimental Data:

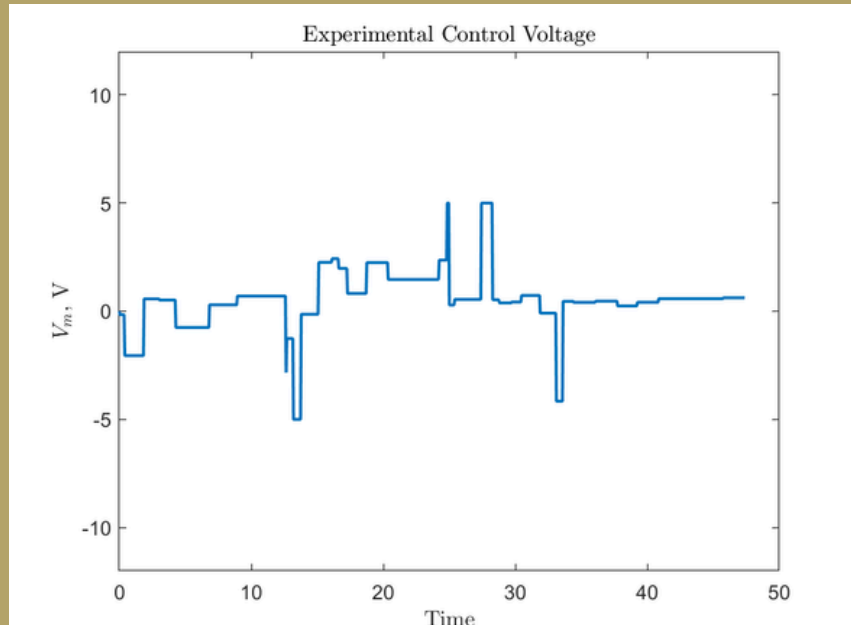


Position of the arm as it tries to stabilize

The system stabilized itself, then experienced an impulse at t=27s. The system then restabilized to the set point. While different than the simulation, the behavior of the system is pretty close to predicted.

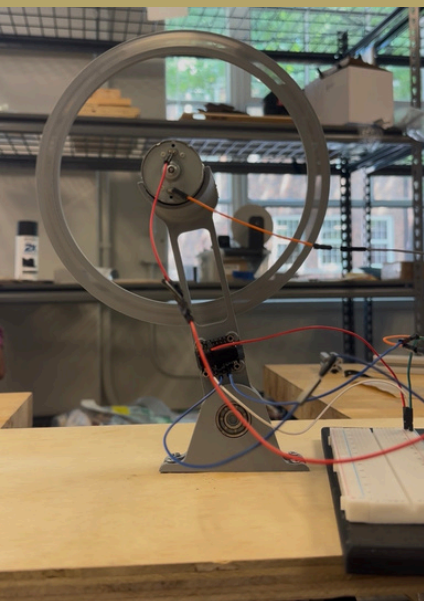
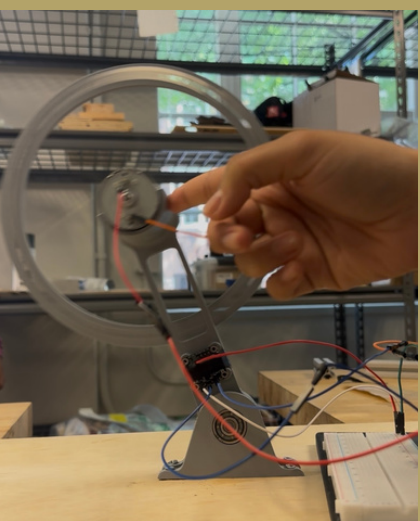


Angular Velocity during the stabilization experiment



Control Effort as the system stabilizes

Stabilization Showcase



CONTROLLER DESIGN

Controller Type

- The LQR method was implemented to design the controller
- This method was selected because the design requirements for system response and control input are specified in terms of limits on excursions from equilibrium (deviations of the pendulum arm from upright position)

Closed-Loop Stability & Gain Tuning

The root locus plot is used to verify the closed-loop stability, as all the poles are in the LHP

Gains were initialized and then tuned via trial and error:

- $K_p = 5$
- $K_i = 1$
- $K_d = 0.5$

Desired Controller Performance Criteria

- Rise time ≤ 1 s
- Settling time ≤ 2 s
- Steady state error: $\Theta = \pm 3^\circ$
- Maximum overshoot: $\Theta = 13^\circ$

CONCLUSION

Findings Summary

- The reaction wheel successfully balanced the inverted pendulum
- Controller criteria was not met but it was close
- System was slow to balance
- SSE was high
- Maximum overshoot was high
- The experimental data differed from the simulated data - potential error in modeling
- Potential errors in IMU calibration

Future Iterations

- A more powerful motor would work faster
- Rotational friction greatly impacts balancing
 - Printing the pieces to be looser fit
 - Using better quality ball bearings
- Better experience with Arduino IDE would allow us to optimize the controller code and improve performance